

Table of Contents

1. Introduction
 - 1.1 Project Overview
 - 1.2 Purpose
 - 1.3 Stakeholders
2. Business Model
3. User Roles
 - 3.1 Super Admin
 - 3.2 Vendor
 - 3.3 Customer
4. Functional Requirements
 - 4.1 Authentication Module
 - 4.2 Vendor Store Management
 - 4.3 Product Management
 - 4.4 Inventory Management
 - 4.5 Product Discovery
 - 4.6 Wishlist Management
 - 4.7 Shopping Cart
 - 4.8 Order Management
 - 4.9 Payment Management
 - 4.10 Reviews & Ratings
 - 4.11 Notification System
 - 4.12 Analytics Dashboard
5. Subscription Plans
6. Non-Functional Requirements
7. Technology Stack
8. Database Entities
9. Future Enhancements
10. Conclusion

1 Introduction

1.1 Project Overview

The **Multi-Tenant E-Commerce Marketplace Platform** is a Software-as-a-Service (SaaS) solution that enables multiple vendors to create and manage independent online stores within a unified marketplace ecosystem. The platform allows customers to browse products from different vendors, place orders, make secure online payments, track shipments, and provide ratings and reviews.

Vendors can manage inventory, process orders, monitor sales, and analyze business performance through dedicated dashboards. The platform administrator manages vendors, subscriptions, commissions, platform analytics, and overall system operations.

1.2 Purpose

The purpose of this system is to provide businesses with an **affordable and scalable** online selling platform without requiring them to build separate e-commerce websites.

- Support multiple vendors simultaneously on a shared infrastructure
- Centralize online selling operations under one unified platform
- Provide secure transactions using industry-standard encryption
- Improve customer shopping experience with intuitive UX
- Generate revenue through subscriptions and commission-based models

1.3 Stakeholders

Stakeholder	Role	Primary Responsibilities
Super Admin	Platform Manager	Manages vendors, commissions, analytics, disputes, and system operations
Vendors	Store Owners	Creates stores, manages products, processes orders, tracks earnings
Customers	End Users	Browses products, places orders, makes payments, writes reviews

2 Business Model

The platform follows a **B2B2C (Business-to-Business-to-Consumer)** marketplace model, connecting vendors with end customers through a centralized digital marketplace.

Business Flow



Revenue Sources

#	Revenue Stream	Description
1	Monthly Subscription Plans	Vendors pay tiered monthly fees for Free, Basic, or Pro plans
2	Commission on Orders	Percentage deducted from each vendor's completed order value
3	Future Promotional Services	Paid advertising, featured listings, and banner placements

3 User Roles

3.1 Super Admin

The Super Admin has full control over the entire platform and is responsible for platform health, vendor management, and financial oversight.

- Manage Vendors — approve, suspend, or remove vendor accounts
- Manage Stores — monitor and configure all vendor stores
- Manage Categories — create and organize product categories
- Configure Commissions — set per-vendor or global commission rates
- Configure Subscription Plans — define pricing tiers and feature sets
- View Platform Analytics — access aggregate sales and revenue dashboards
- Suspend Vendors — restrict access for policy violations
- Handle Disputes — mediate between customers and vendors
- Monitor Payments — track all financial transactions on the platform

3.2 Vendor

Vendors are businesses that sign up to sell products through the marketplace, operating within their subscription plan's limits.

- Create Multiple Stores — launch and brand distinct storefronts
- Manage Products — add, update, and remove product listings
- Manage Inventory — track stock levels and receive low-stock alerts
- Process Orders — confirm, pack, and dispatch customer orders
- View Analytics — review revenue trends and bestselling products
- Manage Discounts — create coupon codes and promotional pricing
- Respond to Reviews — engage with customer feedback publicly
- Track Earnings — monitor payouts and commission deductions

3.3 Customer

Customers are end-users who browse, discover, and purchase products from multiple vendors within a single shopping experience.

- Register / Login — create and access their personal account
- Browse Products — explore the full marketplace catalog
- Search Products — use keywords, filters, and sorting to find items
- Add to Wishlist — save products for future consideration
- Add to Cart — collect items before checkout
- Place Orders — complete secure purchases using integrated payment
- Track Orders — follow real-time order and shipping status
- Rate Products — assign star ratings to purchased items
- Write Reviews — submit detailed product feedback

4 Functional Requirements

4.1 Authentication Module

The authentication module provides secure access control for all user roles, leveraging JWT tokens and role-based access control (RBAC).

Feature	Description
User Registration	New users can create accounts with email verification
User Login	Secure login with hashed password comparison
Password Encryption	All passwords stored using Bcrypt hashing (salt rounds ≥ 10)
Forgot Password	Initiates email-based password recovery flow
Password Reset	Secure tokenized reset link with expiry
JWT Authentication	Stateless session management via signed JSON Web Tokens
Role-Based Access Control	Permissions enforced per role: Admin, Vendor, Customer

4.2 Vendor Store Management

Vendors can create and configure branded storefronts independently, limited by their subscription plan.

- Create Multiple Stores — each store has a unique slug, name, and branding
- Edit Store Information — update name, description, contact details
- Upload Store Logos — cloud-hosted images via Cloudinary integration
- Configure Store Settings — shipping zones, currency, and store policies
- Manage Store Status — toggle store between active, paused, or closed states

4.3 Product Management

Vendors can build rich product catalogs with multi-image support and flexible variant configurations.

Action	Details
Add Products	Create product listings with title, description, price, and category
Update Products	Modify any product attribute without deleting the listing
Delete Products	Soft-delete or permanently remove product entries
Upload Multiple Images	Bulk image upload via Cloudinary with ordering control
Manage Product Variants	Configure variations: Size, Color, Weight, Storage, RAM, Custom

Supported Variant Types: Size • Color • Weight • Storage • RAM • Custom Attributes

4.4 Inventory Management

The system automatically maintains accurate stock counts and prevents overselling across all vendor stores.

- Track Product Stock — real-time quantity tracking per SKU/variant
- Automatically Reduce Stock After Purchase — atomic decrement on order confirmation
- Generate Low Stock Alerts — email/in-app notification when stock falls below threshold
- Prevent Overselling — stock validation before order placement to avoid negative inventory

4.5 Product Discovery

Customers have powerful discovery tools to find exactly what they're looking for across all vendors.

Feature	Description
Search Products	Full-text search across titles, descriptions, and tags
Filter Products	Filter by category, price range, rating, and vendor
Sort Products	Sort by price (asc/desc), rating, newest, and popularity
View Product Details	Rich product page with images, variants, reviews, and vendor info
View Product Reviews	Aggregated rating display with individual review threads

4.6 Wishlist Management

Customers can save items they are interested in for a future purchase, making the shopping experience more convenient.

- Add Products to Wishlist — one-click save from product pages or search results
- Remove Products from Wishlist — manage saved items at any time
- Move Wishlist Items to Cart — direct transfer with quantity selection

4.7 Shopping Cart

A persistent, cross-session shopping cart allows customers to collect items from multiple vendors before checkout.

- Add Products — add single or multiple units from any vendor store
- Update Quantities — increment or decrement item counts inline
- Remove Products — delete individual items or clear the entire cart
- Calculate Total Price — real-time subtotal, discount, tax, and grand total breakdown

4.8 Order Management

The order lifecycle is tracked end-to-end with automated status transitions and notification triggers.

Order Status Flow



Capability	Description
Generate Orders	Order created with unique ID on successful payment
Track Orders	Customer and vendor can monitor status in real time
Update Order Status	Vendors move orders through the workflow stages
Generate Invoices	Downloadable PDF invoices auto-generated on order completion

4.9 Payment Management

All financial transactions are processed through **Stripe Connect**, ensuring PCI-compliant, secure vendor payouts with automatic commission deduction.

- Secure Checkout — SSL-encrypted payment forms via Stripe Elements
- Payment Verification — webhook-based order confirmation for reliability
- Automated Vendor Payouts — funds split and transferred to vendor Stripe accounts
- Commission Deduction — platform fee withheld automatically before payout
- Transaction History — full audit trail accessible to admins and vendors

4.10 Reviews & Ratings

A transparent review system builds trust and helps customers make informed purchase decisions.

Actor	Feature	Details
Customer	Rate Products	1-5 star rating submitted post-delivery only
Customer	Submit Reviews	Text review with optional photos, moderated for spam
Vendor	Reply to Reviews	Public vendor response visible beneath the review
Vendor	View Review Analytics	Aggregate sentiment, rating trends, and flagged reviews

4.11 Notification System

A dual-channel notification system keeps all stakeholders informed of important platform events.

Email Notifications

- Registration — welcome email with account verification link
- Order Confirmation — itemized order summary sent to customer and vendor
- Payment Confirmation — payment receipt with transaction ID
- Shipping Updates — tracking information and estimated delivery date

In-App Notifications

- New Orders — real-time alert for vendors when a new order arrives
- Status Updates — push notification for each order stage transition
- Promotions — targeted alerts for discounts and new vendor launches
- Subscription Reminders — advance warning before plan renewal or expiry

4.12 Analytics Dashboard

Dedicated analytics dashboards provide actionable business intelligence for both vendors and the platform administrator.

Vendor Analytics

- Revenue Reports — daily, weekly, and monthly revenue breakdown
- Orders Overview — total, pending, completed, and cancelled orders
- Best Selling Products — ranked by units sold and revenue generated
- Customer Insights — new vs returning customers, geographic data
- Monthly Sales Trends — line chart of revenue over rolling 12 months

Admin Analytics

- Total Revenue — platform-wide gross and net revenue figures
- Total Vendors — active, pending, and suspended vendor counts
- Total Customers — registered and active customer metrics
- Order Volume — total orders across all vendors and time ranges
- Subscription Revenue — monthly recurring revenue (MRR) by plan tier
- Commission Revenue — total commissions collected per period

5 Subscription Plans

Vendors choose from three tiered subscription plans based on their business size and requirements. Each plan determines the number of stores, products, and feature access available.

Feature	Free Plan	Basic Plan	Pro Plan
Number of Stores	1	3	Unlimited
Products Allowed	20	500	Unlimited
Analytics	Basic	Advanced	Premium
Priority Support	✗	✗	✓
Custom Domain	✗	✗	✓
Promotional Tools	✗	Limited	Full Access

6 Non-Functional Requirements

Category	Requirement	Details
Performance	Response Time	API responses must complete in under 3 seconds
Performance	Search Speed	Product search results delivered within 1 second
Performance	DB Optimization	Indexed queries, pagination, and caching for large datasets
Security	JWT Authentication	All API endpoints protected with signed JWT tokens
Security	Password Hashing	Bcrypt with minimum 10 salt rounds
Security	Helmet Middleware	HTTP security headers applied to all responses
Security	Input Validation	Server-side validation with sanitization on all inputs
Security	Rate Limiting	Request throttling to prevent brute-force and DDoS attacks
Security	CSRF Protection	Anti-CSRF tokens for all state-changing operations
Reliability	Uptime SLA	99.9% availability (max ~8.7 hours downtime/year)
Scalability	Vendors	Support thousands of concurrent vendor accounts
Scalability	Products	Handle millions of product listings efficiently
Maintainability	Architecture	Modular, layered architecture with clean separation of concerns
Maintainability	Code Standards	Enforced ESLint rules, JSDoc comments, and naming conventions
Maintainability	API Docs	OpenAPI/Swagger documentation for all endpoints

7 Technology Stack

The platform is built on a modern JavaScript-based MERN stack with cloud-native services for storage, payments, and email delivery.

Layer	Technology	Purpose
Frontend	React.js	Component-based UI library for building interactive interfaces
Frontend	Redux Toolkit	Centralized state management with simplified Redux patterns
Frontend	Tailwind CSS	Utility-first CSS framework for rapid, responsive styling
Frontend	React Router	Client-side routing for single-page application navigation
Backend	Node.js	Server-side JavaScript runtime for scalable API development
Backend	Express.js	Minimalist web framework for RESTful API construction
Database	MongoDB	NoSQL document database for flexible, schema-less data storage
Database	Mongoose	ODM library providing schema validation and query building
Cloud	Cloudinary	Media storage and transformation CDN for product images
Cloud	Stripe Connect	Payment processing with multi-party split payment support
Cloud	Nodemailer	SMTP-based transactional email delivery service
Security	JWT	Stateless authentication via signed JSON Web Tokens
Security	Bcrypt	Adaptive password hashing with configurable work factor
Security	Helmet.js	Express middleware setting secure HTTP response headers
Deployment	Vercel	Frontend hosting with CI/CD integration and edge CDN
Deployment	AWS / Render	Backend API hosting with auto-scaling and managed environments

8 Database Entities

The platform's MongoDB database comprises the following core collections (entities). Each entity maps to a Mongoose schema with defined fields, validations, and relationships.

#	Entity	Description
1	Users	Stores customer, vendor, and admin account data including authentication credentials, roles, and profile info.
2	Stores	Represents vendor storefronts with branding, settings, status, and subscription linkage.
3	Products	Core product catalog with pricing, images, description, category, and vendor association.
4	Product Variants	Stores variant-level data (size, color, etc.) with individual stock counts and pricing overrides.
5	Categories	Hierarchical product taxonomy managed by the admin for consistent classification.

#	Entity	Description
6	Cart	Temporary per-user container of selected products and quantities before checkout.
7	Wishlist	Saved product references per user, persisted across sessions.
8	Orders	Complete order record including items, quantities, prices, status, and shipping info.
9	Payments	Payment transaction log with Stripe charge IDs, status, and amount breakdowns.
10	Reviews	Customer text reviews linked to specific products and orders, with vendor reply support.
11	Ratings	Numeric star ratings stored separately for efficient aggregation and display.
12	Notifications	In-app notification records with type, message, read status, and recipient.
13	Subscriptions	Vendor subscription plan details, billing cycle, and renewal dates.
14	Vendor Payouts	Records of funds transferred to vendors after commission deduction.
15	Analytics	Pre-aggregated metrics snapshots for dashboard performance optimization.

9 Future Enhancements

The following features are planned for future platform releases, expanding capabilities and market reach:

Enhancement	Description	Priority
Mobile Applications	Native iOS and Android apps for customers and vendors	High
AI Product Recommendations	ML-based personalized product suggestions using purchase history	High
Multi-Language Support	Internationalization (i18n) for global vendor and customer base	Medium
Multi-Currency Support	Real-time currency conversion and localized pricing	Medium
Loyalty Program	Points-based reward system for repeat customers	Medium
Vendor Advertising System	Self-serve paid promotional placements and featured listings	Medium
Live Chat Support	Real-time customer-to-vendor messaging within the marketplace	Low
Advanced Reporting	Custom report builder with CSV/PDF export for vendors and admin	Low

10 Conclusion

The **Multi-Tenant E-Commerce Marketplace Platform** is a comprehensive, scalable SaaS-based marketplace solution that addresses the needs of modern digital commerce. By enabling multiple vendors to operate independent storefronts within a shared ecosystem, the platform reduces the barrier to entry for businesses seeking an online presence.

The platform's architecture — built on the MERN stack with Stripe Connect, Cloudinary, and JWT-based security — ensures performance, reliability, and extensibility. The tiered subscription model and commission-based revenue strategy create a sustainable business model for the platform while remaining affordable for vendors at all scales.

With a clear roadmap for future enhancements including mobile applications, AI recommendations, and multi-language support, the platform is positioned for long-term growth and international market expansion. This SRS document serves as the definitive technical blueprint for the design, development, and deployment of the platform.

Key Takeaways:

- Multi-tenant architecture supports thousands of vendors simultaneously
- Secure payment processing via Stripe Connect with automated commission deduction
- Comprehensive analytics for both vendors and platform administrators
- 99.9% uptime SLA with scalable cloud-native infrastructure
- Extensible design ready for AI, mobile, and global market features